

## Relationship of Glymphatic System to Sleep and Neurodegenerative Disorders

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### Abstract

Neurodegenerative disorders; such as, Alzheimer's and Parkinson's diseases; are a pressing public health concern with significant implications for the patient's families, direct care givers and society. Glymphatic system, which can be seen as a waste disposal system in the brain, has been implicated in neurodegenerative disorders. Critically, recently studies have also shown that glymphatic flow is affected by sleep. Hence, this minireview aims to elucidate the relationship between glymphatic system, sleep, and neurodegenerative disorders; as a basis for further research into possible preventive and/or early corrective measures for neurodegenerative disorders.

**Keywords:** Glymphatic System (GS); Cerebrospinal Fluid (CSF); Central Nerverous System (CNS); Alzheimer's disease (AD); Parkinson's disease (PD)

### Introduction

The Glymphatic System (GS) was discovered in 2012 by Iliff and Nedergaard [1]. It refers to the cerebrospinal fluid (CSF) and GS creating a brain perivascular system that uses arterial pulsations [2], which removes metabolic and neurotoxic waste, and circulating essential nutrients with signaling ligands in the central nerverous system (CNS) [3]. During sleep, noradrenergic system which release norepinephrine is reduce significantly and result in dilation of the interstitial space within the brain by up to 60% [4], the dilation increase the permeability of the extracellular space which enable faster clearing of metabolic waste [5], in contrast GS function is severely suppressed during awake [6]. GS is influence by the sleep-wake state and this highlight the importance of sleep in maintaining brain homeostasis and metabolic waste removal [7]. Current studies have term CSF circulation and GS function as interchangeably and reflect interdependent but distinct dynamics [8]. More importantly, dysfunction of GS has been closely identified as root cause in pathogenesis and aggravation which turn into various neurodegenerative diseases such as Alzheimer's disease (AD) and Parkinson's disease (PD) [9]. This minireview seeks to understand the relationship between GS, sleep, and neurodegenerative disorders.

### Function of glymphatic system

Primary function of GS is to helps the brain create a stable cerebral homeostasis using CSF ventricular system within the brain via perivascular tunnels in the brain which faciliate the movement of CSF and ISF, which clears metabolic waste and neurotoxic protein from the brain, while supplying important nutrients and signaling molecules within the brain [8].

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CSF is produced in the choroid plexus located in the ventricles between the 3<sup>rd</sup> and 4<sup>th</sup> ventricles of the brain and consists of epithelium cells, movements of CSF is through the ventricles system inside the cisterns and subarachnoid space. CSF is absorbed into arachnoid villi and meningeal lymphatic vasculature (MLVs) [10]. During circadian rhythm, CSF creation and circulation are spike up significantly [11]. CSF circulation from subarachnoid space to Periarterial spaces (PS) surrounding the brain parenchyma and form part of the GS function but the final efflux process is still obscure but assume to include waste clearance of solutes along VRS and meningeal lymphatic vessels [10]. Astrocytic aquaporin-4 (AQP4) water channels are critical for GS as it aid fluid circulation within periarterial spaces (entry of fluid of CSF into brain parenchyma) and perivenous spaces (exit of interstitial fluid ISF) waste clearance system pathway and is regulated by circadian rhythm [7].

Astrocytes are helper cells that support neurons which relay signals in the interconnected network in the brains, it has extensions named astrocytic end feet which enwrap cerebral capillaries in the brain and spinal cord, they create a pathway and allow fluid to rapidly transport through the brains, clearing metabolic waste from the parenchyma into the GS [12].

GS is essential in the clearance of alpha-synuclein, a type of presynaptic protein which facilitate release of neurotransmitters [3] at synaptic to enable communication between nerve cells [13]. GS is dependent on CSF circulation in the brain but other hypothesis of waste clearance systems, which remove beta-amyloid (protein associated with Alzheimer's disease) across the blood-brain barrier have been propose. This implies that the CSF circulation system is not totally exclusive to GS. Instead, CSF circulation and waste clearance can be via other mechanism such as protein or solutes movement via blood thru CSF barrier which is different from GS. Glymphatic flow comprises of CSF penetrating PS in the brain which penetrate into tissue then mixed with ISF then transport the solutes into PS and discharged from the brain. Hence, GS allows mass fluid circulation which aid with brain homeostasis and is active during sleep [14].

### Role of sleep in glymphatic system

Good sleep has always been essential to one's physical and mental wellness, neurocognitive performance, euthymia, circulatory system, cerebrovascular and metabolism health [15]. A study on effects of sleep deprivation (SD) stated that SD has established the adverse effect on one's health with increase hypertension, coronary heart issues and when compared between male and female. SD has a bigger negative impact on male; espically on obesity and hypertension, persons under 65 years old are more negatively affected by SD than above 65, mainly in cardiovascular disease and overall mortality [16]. Researchers wanted to determine if SD could lead to neurodegenerative disorders, targeted proteins closely associated with neurodegenerative brain diseases; such as, beta-amyloid, alpha-synuclein, and tau protein; are continuously release as metabolic waste in normal neuronal activities and are accumulated within the interstitium space which surround the brain. When excessive amount are accumulated or insufficiently cleared, the waste can accumulate and release neurotoxin, which are responsible for disorders like AD and PD.

Clinical trials using mice demonstrated that during sleep, GS cleared beta-amyloid more efficiently, and the clearance rate almost double during sleep. In another trial, mice which are genetically designed to lacked AQP4 in astrocytes exhibited significant impaired glymphatic activity, clearance of beta-amyloid were 40% lesser than control mice [17]. The proportion of beta-amyloid removed during the trial was prominent and it questioned the accepted concept that brain cells process the metabolic waste internally via cell degradation method called ubiquitnation and autophagy. Clinical results showed that brain flush out significant amount of metabolic waste including beta-amyloid via GS, with degradation happening elsewhere within the body system. The clinical findings supported the thesis that the brain actively flush out metabolic waste during sleep [18]. This is supported by another study showing that mice have higher levels of beta-amyloid when awake then asleep and SD exacerbate amyloid plaque accumulation in mice which are genetically modified to bioaccumulation in excess [6].

### Linking glymphatic system to neurodegenerative disorders

GS has been a critical part in maintaining homeostasis of the brain by assisting the clearance of neurotoxic proteins and metabolic waste [3]. Dysfunction of glymphatic system due to sleep deprivation, aging or vascular impairment will result in building-up of neurotoxic in the GS [19]. Dysfunction has lead to an accumulation of tau and beta-amyloid proteins, which have been linked to cognitive decline, symptoms associated to dementia and AD [8,20,21]. Aging, hypertension, and reduce cardiac output will affect the CSF inflow in the GS as it is primarily driven by arterial pulsation [22]. When the clearance rates are affected, metabolic waste accumulation will start to form and an early buildup of beta-amyloid linking to AD and vascular dementia would start to form [23,24].

Using protein TDP-43 on mouse, to model amyotrophic lateral sclerosis (ALS; a type of degenerative neurodegenerative disorder that target the motor neurons), results shows that TDP-43 affecting motoneurons degeneration [25-28]. ALS progressively deteriorate neurons between the brain and spinal cord, and this cause loss of voluntary movement and potential paralysis, especially when GS dysfunction failed to clear the neurotoxic protein [26,27,29,30].

Parenchymal flow a part of GS and is dependent on factors; such as, arterial pulsations, AQP4 channels, and sleep state [6,31]. When there is failure of parenchymal flow, cause by weak arterial pulsations caused by aging, hypertension and reduce slow-wave sleep patterns [6,10,24,26,32]. There may be an accumulation of metabolic waste within the interstitial spaces [33], combine retention of beta-amyloid and tau will synergistically in triggering AD pathogenesis [34]; while alpha-synuclein build up increases formation of lewy bodies, which would to neurodegenerative pathway called synucleinopathies leading to PD and dementia [20,35].

### Conclusion

An average person spent about one-third of their life sleeping; hence, sleep is an important area of study. GS, a waste clearance system in the brain, has been shown to increase activity during sleep. Importantly, GS is potential factor in the etiopathogenesis for various neurodegenerative diseases; such as, dementia, cognitive decline, AD, and PD. Hence, improving GS efficiency and clearance functions may be important preventive measures in neurodegenerative disorders.

### Conflict of Interest

The authors declare no conflict of interest.

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